Machine Learning

Lecture 12

Probabilities (recap)

Probabilities (Recap)

- $P(X = 0) = ?$
- $P(Y = 3) = ?$
- $P(X = 1, Y = 2) = ?$
- $P(Y = 2, X = 1) = ?$
- $P(X = 1 \mid Y = 2) = ?$
- $P(Y = 2 \mid X = 1) = ?$

X	Y
0	0
0	1
1	0
1	2
2	3
2	0
2	3
1	3
1	2
0	3
0	2
0	0

Probabilities (Recap)

- $P(X = 0) =$ **5/12**
- $P(Y = 3) =$ **4/12**
- $P(X = 1, Y = 2) =$ **2/12**
- $P(Y = 2, X = 1) =$ **2/12**
- $P(X = 1 \mid Y = 2) =$ **2/3**
- $P(Y = 2 \mid X = 1) =$ **2/4**

X	Y
0	0
0	1
1	0
1	2
2	3
2	0
2	3
1	3
1	2
0	3
0	2
0	0

Probabilities (Recap)

if all events are independent

- P (A , B) = P(A) * P (B)
- P (A , B , C) = P(A) * P (B) * P(C)
- P (A , B , C, D) = P(A) * P (B) * P(C) * P(D)
- $P(A_1,A_2, . . ., A_n) = P(A_1) * P (A_2) * ..... * P(A_n)$

- $P(A|B) = \frac{P(A\ and\ B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$

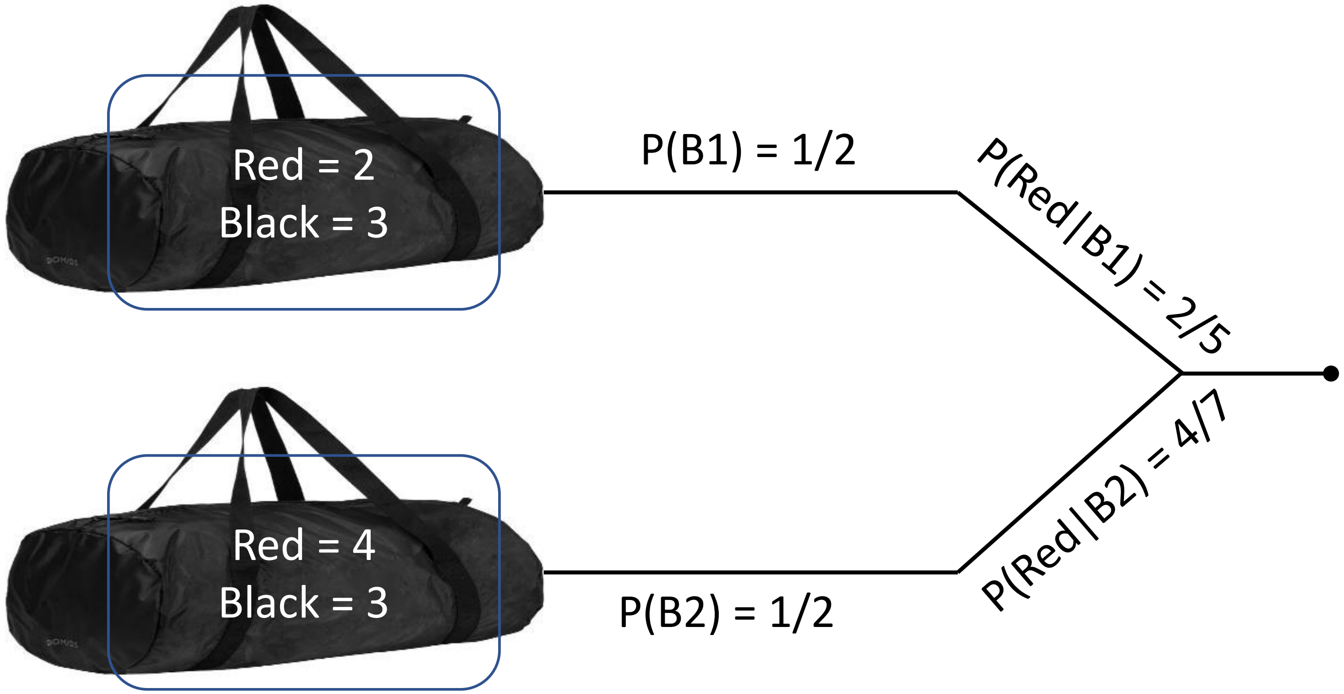
Conditionally independent

- P (A , B | C) = P(A | C) * P (B | C)
- P (A , B , C | D) = P(A | D) * P (B | D) * P (C | D)
- $P(A_1,A_2, . . ., A_n \mid Z) = P(A_1 \mid Z) * P (A_2 \mid D) * P (A_n \mid D)$

Understanding Baye’s Theorem

- Bag 1 contains 2 red 3 black balls. Bag 2 contains 4 red and 3 black balls. One ball is drawn from one of these bags and its red. Find the probably that it is drawn from Bag 1.

One ball is drawn from one of these bags. What is the probability that the ball is red?

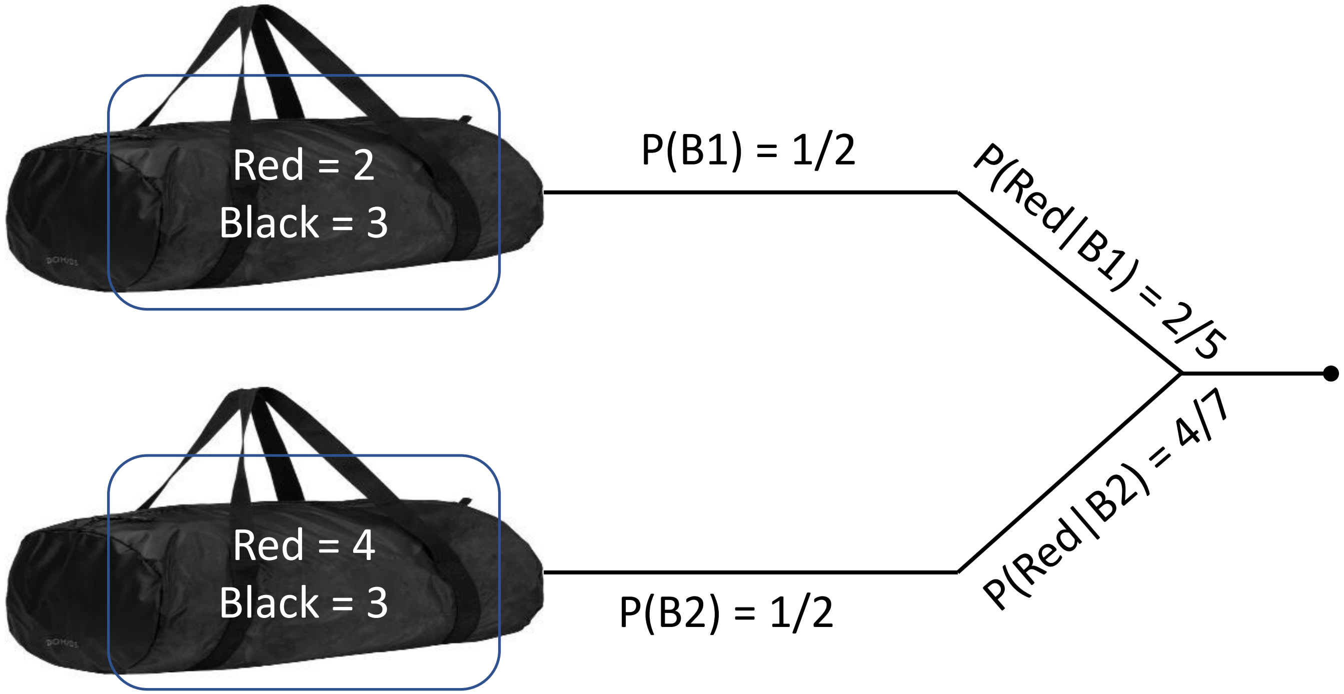


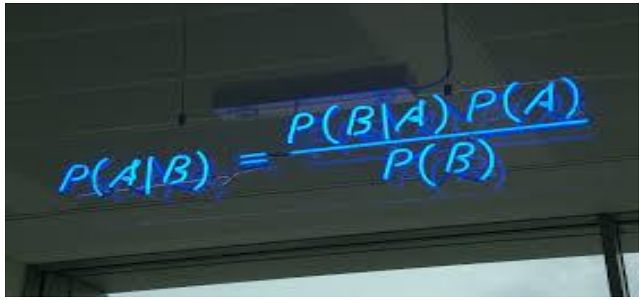
P(Red) = P(B1)×P(Red|B1) + P(B2)×P(Red|B2)

$$= \frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{4}{7} = 0.78$$

Understanding Baye’s Theorem

- Bag 1 contains 2 red 3 black balls. Bag 2 contains 4 red and 3 black balls. One ball is drawn from one of these bags and its red. Find the probably that it is drawn from Bag 1. i.e. P(B1|R) = ?





$$\mathbf{P(B1|Red)} = \frac{\boldsymbol{P(Red|B1) \times P(B1)}}{\boldsymbol{P(Red)}}$$

$$= \frac{\frac{2}{5} \times \frac{1}{2}}{\boldsymbol{P(Red)}}$$

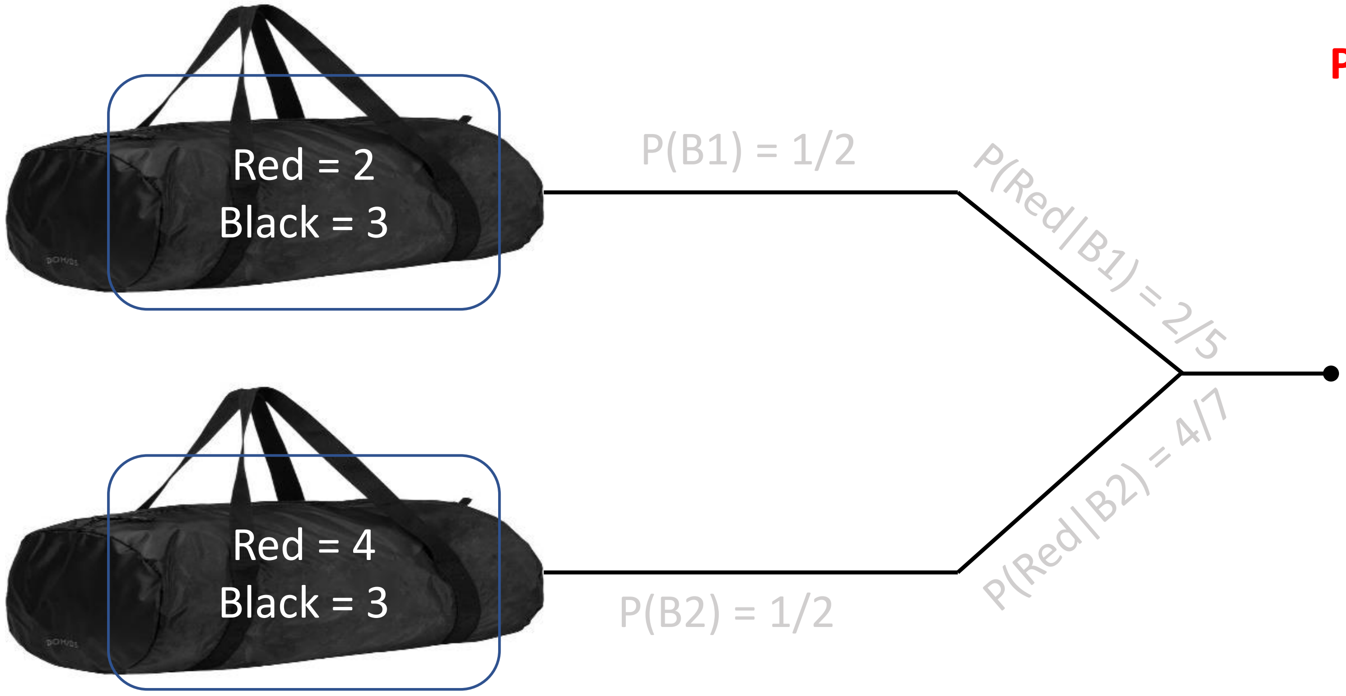
Understanding Baye's Theorem

- Bag 1 contains 2 red 3 black balls. Bag 2 contains 4 red and 3 black balls. One ball is drawn from one of these bags and its red. Find the probably that it is drawn from Bag 1. i.e. P(B1|R) = ?

P(Red) = ?

P(Red) = P(B1)×P(Red|B1) + P(B2)×P(Red|B2)

$$= \frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{4}{7} = 0.78$$



$$\mathbf{P(B1|Red)} = \frac{P(Red|B1) \times P(B1)}{P(Red)}$$

$$= \frac{0.4 \times 0.5}{0.78}$$

Understanding Baye’s Theorem

- It calculates the probability of a class C, given a set of features X.

$$P(c|X) = \frac{P(X|c) \cdot P(c)}{P(X)}$$

$$P(X|c) = P(x_1, x_2, \ldots, x_n|c)$$

$$P(x_1, \ldots, x_n|c) = P(x_1|c) \bullet P(x_2|c) \bullet P(x_3|c) \bullet \ldots \bullet P(x_n|c)$$

Naïve Bayes Classifier

(Classification Algorithms)

Naïve Baye’s Classifier

- Naïve Bayes is a supervised machine learning algorithm used for classification based on Bayes' Theorem.
- The *Naïve* Part:

> It assumes that all input features (X_1, X_2, ...) are conditionally independent of each other, given the class (C).

- It assumes that knowing the value of one feature tells you nothing about the value of another feature, if you already know the class. For example, in spam detection, it assumes that the presence of the word “free” is independent of the presence of the word “education”, given that the email is spam (or not spam).

$$\underbrace{P(c \mid x)}_{\text{Posterior Probability}} = \frac{\overbrace{P(x \mid c)}^{\text{Likelihood}}\,\overbrace{P(c)}^{\text{Class Prior Probability}}}{\underbrace{P(x)}_{\text{Predictor Prior Probability}}}$$

$$P(c \mid \mathbf{X}) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(c \mid x)$$

MAP is "maximum a posteriori" = most likely class (e.g. when you have more than 3 classes C_{MAP} = MAX (P(C1|x), prob (C2|x), Prob(C3|x)).

$$= \underset{c \in C}{\operatorname{argmax}} \frac{P(x \mid c)P(c)}{P(x)}$$

Bayes Rule

$$= \underset{c \in C}{\operatorname{argmax}} P(x \mid c)P(c)$$

Dropping the denominator

Example

Person	Cough (Yes/No)	Flu (Yes/No)	Fever (Yes/No)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

- **Step 1: Prior Probability**

  P (Fever: Yes) = 7/10 = 0.7

  P(Fever: No) = 3/10 = 0.3

- **Step 2: Conditional Probability**

	Yes	No
Cough	P(Cough|Yes) = 4/7	P(Cough|No) = 2/3
Flu	P(Flu|Yes) = 3/7	P(Flu|No) = 2/3

Given a Person(Cough, Flu), predict fever.

P (Yes|Cough, Flu) = P(Cough|Yes) × P(Flu|Yes) × P(Yes)

= 4/7 × 3/7 × 7/10 = **0.17**

P(No|Cough, Flu) = P(Cough|No) × P(Flu|No) × P(No)

= 2/3 × 2/3 × 3/10 = **0.13**

Prediction: Fever = Yes.

Example

X1	X2	X3	X4	X5	X6	Class
1	0	0	1	1	1	Yes
1	1	0	1	1	1	Yes
0	1	0	0	0	0	Yes
0	0	1	0	0	1	No
1	1	0	1	1	0	No
0	1	0	1	1	0	No

- Testing Data

1, 1, 0, 0, 0, 1   YES

1, 0, 1, 1, 1, 1   YES

- **Step 1: Prior Probability**

P (Class: Yes) = 3/6 = 0.5

P(Class: No) = 3/6 = 0.5

- **Step 2: Conditional Probability**

P(x1=1|y) =	P(x1=0|y)=
P(x1=1|n) =	P(x1=0|n) =
P(x2=1|y) =	P(x2=0|y) =
P(x2=1|n) =	P(x2=0|n) =
P(x3=1|y) =	P(x3=0|y) =
P(x3=1|n) =	P(x3=0|n) =
P(x4=1|y) =	P(x4=0|y) =
P(x4=1|n) =	P(x4=0|n) =
P(x5=1|y) =	P(x5=0|y) =
P(x5=1|n) =	P(x5=0|n) =
P(x6=1|y) =	P(x6=0|y) =
P(x6=1|n) =	P(x6=0|n) =

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

- Testing Data

$T^{(1)}$ = 1, 1, 0, 0, 0, 1 y

$T^{(2)}$ = 1, 0, 1, 1, 1, 1 y

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

- Testing Data

$T^{(1)}$: 1, 1, 0, 0, 0, 1 y

$T^{(2)}$ = 1, 0, 1, 1, 1, 1 y

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

- $T^{(1)}$: 1, 1, 0, 0, 0, 1
- $P(y|T^{(1)}) = P(X1=1|y) \times P(X2=1|y) \times P(X3=0|y) \times P(X4=0|y) \times P(X5=0|y) \times P(X6=1|y) \times P(y)$
- $P(y|T^{(1)}) = 2/3 \times 2/3 \times 1 \times 1/3 \times 1/3 \times 2/3 \times 1/2$
- $P(y|T^{(1)}) = 0.015$

- $P(n|T^{(1)}) = P(X1=1|n) \times P(X2=1|n) \times P(X3=0|n) \times P(X4=0|n) \times P(X5=0|n) \times P(X6=1|n) \times P(n)$
- $P(n|T^{(1)}) = 1/3 \times 2/3 \times 2/3 \times 1/3 \times 1/3 \times 1/3 \times 1/2$
- $P(n|T^{(1)}) = 0.0025$

- Prediction: Class = y.

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

- Testing Data

$t^{(1)}$: 1, 1, 0, 0, 0, 1 y

$t^{(2)}$: 1, 0, 1, 1, 1, 1 y

- $P(t^{(2)} | c=y) =$

- $P(t^{(2)} | c=n) =$

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =2/3 P(x1=0|y) =1/3

P(x1=1|n) = 1/3 P(x1=0|n) = 2/3

P(x2=1|y) = 2/3 P(x2=0|y) = 1/3

P(x2=1|n) = 2/3 P(x2=0|n) = 1/3

P(x3=1|y) = 0 P(x3=0|y) = 1

P(x3=1|n) = 1/3 P(x3=0|n) = 2/3

P(x4=1|y) = 2/3 P(x4=0|y) = 1/3

P(x4=1|n) = 2/3 P(x4=0|n) = 1/3

P(x5=1|y) = 2/3 P(x5=0|y) = 1/3

P(x5=1|n) = 2/3 P(x5=0|n) = 1/3

P(x6=1|y) = 2/3 P(x6=0|y) = 1/3

P(x6=1|n) = 1/3 P(x6=0|n) = 2/3

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

Problem of zero probability…

- So, how to deal with this problem?

Naïve Bayes Classifier (smoothing)

Naïve Bayes Classifier (Smoothing)

- Laplace smoothing/additive smoothing is a smoothing technique that handles the problem of zero probability in Naïve Bayes.
- A small-sample correction, or **pseudo-count**, will be incorporated in every probability estimate.
- Consequently, no probability will be zero.
- This is a way of regularizing Naive Bayes, and when the pseudo-count is zero, it is called Laplace smoothing.
- In general cases it is often called **Lidstone smoothing**.

Laplacian Smoothing

$$P(F = v \mid Y = y) = \frac{\#\{F = v, Y = y\} + 1}{n_y + \underbrace{|V_F|}_{\text{Total distinct values of feature F}}}$$

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = P(n) =

P(x1=1|y) = P(x1=0|y)=

P(x1=1|n) = P(x1=0|n) =

P(x2=1|y) = P(x2=0|y) =

P(x2=1|n) = P(x2=0|n) =

P(x3=1|y) = P(x3=0|y) =

P(x3=1|n) = P(x3=0|n) =

P(x4=1|y) = P(x4=0|y) =

P(x4=1|n) = P(x4=0|n) =

P(x5=1|y) = P(x5=0|y) =

P(x5=1|n) = P(x5=0|n) =

P(x6=1|y) = P(x6=0|y) =

P(x6=1|n) = P(x6=0|n) =

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$

- Training Data

x1	x2	x3	x4	x5	x6	Class
1	0	0	1	1	1	y
1	1	0	1	1	1	y
0	1	0	0	0	0	y
0	0	1	0	0	1	n
1	1	0	1	1	0	n
0	1	0	1	1	0	n

P(y) = 1/2 P(n) = 1/2

P(x1=1|y) =3/5 P(x1=0|y) =2/3

P(x1=1|n) = 2/5 P(x1=0|n) = 3/5

P(x2=1|y) = 3/5 P(x2=0|y) = 2/5

P(x2=1|n) = 3/5 P(x2=0|n) = 2/5

P(x3=1|y) = 1/5 P(x3=0|y) = 4/5

P(x3=1|n) = 2/5 P(x3=0|n) = 3/5

P(x4=1|y) = 3/5 P(x4=0|y) = 2/5

P(x4=1|n) = 3/5 P(x4=0|n) = 2/5

P(x5=1|y) = 3/5 P(x5=0|y) = 2/5

P(x5=1|n) = 3/5 P(x5=0|n) = 2/5

P(x6=1|y) = 3/5 P(x6=0|y) = 2/5

P(x6=1|n) = 2/5 P(x6=0|n) = 3/5

- Testing Data

1, 1, 0, 0, 0, 1 y

1, 0, 1, 1, 1, 1 y

$$\underset{c \in C}{\operatorname{argmax}} P(x \mid c) P(c)$$